1(i)

Given:
$f(x) = ((x^2 + 1) / (x^2 - 1))^{(1/3)}$

$u(x) = x^2$
$v(t) = (t + 1) / (t - 1)$
$w(s) = s^{(1/3)}$

Then:
$f(x) = w(v(u(x)))$

1(ii)

$x^2 - 1 \neq 0$
$x \neq -1, 1$

Domain:
(-infinity, -1) U (-1, 1) U (1, infinity)

Range:

Let $t = x^2$, so $t \geq 0$ and $t \neq 1$.
$R = (t + 1) / (t - 1)$

As t varies over [0, infinity) except 1, R takes values:
(-infinity, -1] U (1, infinity)

Then f(x) = cube_root(R). Cube root does't change which values are possible, just cube roots them, so the range is:

(-infinity, -1] U (1, infinity)

2(i)

cos(pi x) is between -1 and 1
cos(pi x) - 1 is between -2 and 0
|cos(pi x) - 1| is between 0 and 2
2 - 4|cos(pi x) - 1| is between 2 - 8 and 2

Range:
[-6, 2]

2(ii)

g(x) = cos(x)

cos(pi x) is between -1 and 1.
So cos(pi x) + 1 is between 0 and 2, so it's never negative
So |cos(pi x) + 1| = cos(pi x) + 1.

Then:
f(x) = 2 - 4(cos(pi x) + 1)
f(x) = -4cos(pi x) - 2

so:
Amplitude: 4
Midline: y = -2
Period: 2
Range: [-6, 2]

My six plots:

1. Plot 1: g(x) = cos(x)

2. Plot 2: g1(x) = cos(pi x)
   period becomes 2

3. Plot 3: g2(x) = cos(pi x) + 1
   midline becomes y = 1

4. Plot 4: g3(x) = |cos(pi x) + 1|
   it's always >= 0

5. Plot 5: g4(x) = -4|cos(pi x) + 1|

6. Plot 6: g5(x) = 2 - 4|cos(pi x) + 1|
   shift up by 2

3(i)

P(t) = P(0) * 2^(t/5)

3(ii)

$10000 = P(0) * 2^{(20/5)}$
$10000 = P(0) * 16$
$P(0) = 625$

4(i)

$\log_4(48) - \log_4(3) = \log_4(48/3) = \log_4(16) = 2$

4(ii)

$10^{(-(1/2) \log_{10}(49))}$
$= (10^{(\log_{10}(49))})^{(-1/2)}$
$= 49^{(-1/2)}$
$= 1/7$

4(iii)

$\log_a(b) = \ln(b)/\ln(a)$
$\log_b(a) = \ln(a)/\ln(b)$

$1 / \log_b(a) = 1 / (\ln(a)/\ln(b)) = \ln(b)/\ln(a) = \log_a(b)$

4(iv)

$\arcsin(-1/2) = -\pi/6$
$\tan(-\pi/6) = -1/\sqrt{3} = -\sqrt{3}/3$

4(v)

Let $f(x) = (1 - e^x)/(1 + e^x)$

$f(-x) = (1 - e^{(-x)})/(1 + e^{(-x)})$
Multiply numerator and denominator by e^x:

$f(-x) = (e^x - 1)/(e^x + 1)$
$= -(1 - e^x)/(1 + e^x)$
$= -f(x)$

So f is odd.